MSc Artificial Intelligence and Data Science

A Sensorless Markerless Screening Framework for Identifying Kinematic Patterns Associated with Elevated Injury Risk Across Multiple Athletic Exercises

Shubham

Solent University, Southampton

Faculty of Business, Law and Digital Technologies

Supervisor: Dr Raza Hasan

Module: COM726 — Dissertation (Research Project)

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Declaration of Authorship

I declare that this dissertation is my own original work, carried out under the supervision of Dr Raza Hasan, and that all sources of information and material used have been appropriately acknowledged and referenced. This work has not been submitted for any other award.

Signed: ____________________ Date: ______________

Acknowledgements

Abstract

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Acronyms

LoA	Limits of Agreement
LOSO	Leave-One-Subject-Out
LSTM	Long Short-Term Memory
XAI	Explainable Artificial Intelligence

1. Introduction

This harness build cites (Placeholder and Example, 2024) to validate the bibliography pipeline.

2. Background and Literature Review

3. Methodology

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References

A. Placeholder and B. Example. Replace this file with the aligned 16-entry references.bib. *Journal of Build Validation*, 1:1–2, 2024.

A. Data Ethics

A.1 Institutional Ethical Approval

This project received ethical approval from the Solent University Ethics Release Committee prior to commencement of the research.

ERC reference:	`[INSERT ERC REFERENCE]`
Date of approval:	`[INSERT DATE]`
Principal investigator:	Shubham
Supervisor:	Dr Raza Hasan
Faculty:	Business, Law and Digital Technologies
Level:	Postgraduate
Unit code:	COM726

A.2 Scope of Data Collection

No primary human data were collected for this project. All kinematic data analysed in this dissertation are secondary use of three publicly released research datasets, each of which was collected by its originating investigators under their own institutional ethical governance. The research therefore involved no recruitment of participants, no intervention, no collection of identifiable personal data, and no contact with human subjects by the present investigator.

One deviation from this position was considered and rejected during the design phase. A self-recorded cohort of five volunteers was evaluated as a potential supplementary validation set and set aside, on the grounds that recruiting participants and recording video of them would have constituted primary human-subjects data collection requiring a separate and more substantial ethical review than the secondary-use position described here. A single self-recorded video of the investigator is used solely to illustrate the demonstration interface; under the Ethics Release Checklist this does not constitute human participation, which is defined as involving participants *other than* the investigator.

A.3 Dataset Provenance and Licensing

Dataset	Source	Ethical basis and terms of use
REHAB24-6	`[CITE KEY]`	Physical rehabilitation exercise dataset, collected under the originating institution's ethical approval and released for research use. Terms: `[VERIFY LICENCE]`. Used here for the squat and lunge cohorts.
OpenCap	Uhlrich et al. (2023), *PLOS Computational Biology* `[CITE KEY]`	Markerless motion capture validation data released under an IRB-approved protocol by the originating investigators. Terms: `[VERIFY LICENCE]`. Used here as the ground-truth validation cohort for drop-jump.
Penn Action	Zhang, Zhu and Derpanis (2013), *ICCV* `[CITE KEY]`	Public academic dataset of human action video, released for research use. Terms: `[VERIFY LICENCE]`. Used here for the exploratory single-repetition squat subset.

All three datasets are cited in the References in the same manner as any other scholarly source, and their use in this work is consistent with the research purposes for which they were released.

A.4 Data Handling

No personally identifiable information was processed at any stage. Subject identifiers throughout this dissertation are those assigned by the originating dataset providers and carry no link to any individual known to the investigator. Derived kinematic outputs, intermediate artefacts and analysis code are retained in the project repository (Appendix G); no raw video is redistributed, and each dataset remains obtainable only from its original provider under that provider's own terms.

B. Uncertainty Framework Derivation

C. LOSO Fold-by-Fold Results

D. XAI Verification Parameters

E. Bland–Altman Biomarker Interpretation

F. Data Provenance

G. GitHub Repository